Surrogate-Guided Atomic Discovery Finds 42% More Near-Hull Materials Than Random Search

shivanshu sirohi and NeuriCo

Abstract

Autonomous materials discovery promises to replace blind trial-and-error with structured search, but progress depends on whether learned models can allocate scarce validation budget better than random selection. We examine this question in a reproducible offline benchmark using real MATERIALS PROJECT energy-above-hull labels as an oracle. Our loop starts from 400 random observations, trains a leakage-controlled MLP surrogate on composition and coarse cell descriptors, selects 250 additional candidates per round, and repeats for eight rounds under a fixed 2,400-candidate validation budget. We compare random search, uncertainty sampling, greedy exploitation, upper-confidence-bound selection, and diversity-aware upper-confidence-bound selection across five matched seeds. The best policy, DIVERSE-UCB, found 1,933.2 stable or near-hull candidates on average, compared with 1,364.8 for random search, a gain of 568.4 discoveries with a 95% confidence interval of [524.0, 612.8] and Holm-adjusted $p = 1.05 \times 10^{-5}$. Greedy and standard UCB also strongly outperformed random, while pure uncertainty sampling found fewer stable candidates than random. A composition-only transfer check on GNOME R2SCAN summaries improved top-100 on-hull enrichment from 0.8735 to 0.970. These results support a narrow operational claim: surrogate-guided atomic triage can reduce wasted validation effort, but it does not by itself establish new physical materials without higher-fidelity calculation, novelty audit, and synthesis evidence.

1 Introduction

Materials discovery is a search problem with expensive feedback. A practical system cannot validate every possible atomic combination with density functional theory (DFT), synthesis, and characterization. It must decide which candidates deserve validation, learn from the outcomes, and improve the next set of choices. This is the operational core of AI-driven invention in inorganic materials: not a claim that a model has created a finished material, but a test of whether structured selection beats trial-and-error under the same budget.

The recent materials literature shows why this question matters. Large-scale systems such as GNOME use graph networks and active learning to expand the set of predicted stable crystals Merchant et al. [2023]. Benchmarks such as MATBENCH DISCOVERY ask whether models rank stability in the way discovery actually needs Riebesell et al. [2023]. Generative models such as MATTERGEN propose new crystal structures Zeni et al. [2023], and universal atomistic models such as CHGNET, MACE-MP, M3GNET, and OMAT24-derived models make fast screening increasingly practical Deng et al. [2023], Batatia et al. [2024], Chen and Ong [2022], Barroso-Luque et al. [2024]. On the other hand, autonomous-lab results also show that prediction and experiment interpretation must be audited carefully before making novelty claims Szymanski et al. [2023], Nature [2026].

What is still missing? Many systems generate or screen candidates at large scale, but fewer compact benchmarks isolate the decision problem: given a fixed validation budget and a hidden stability oracle, how much more stable material can a structured loop find than random search? This dis-

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tinction matters because a system that only improves model uncertainty may not improve discovery yield. Conversely, a simple model can be useful if it ranks the right candidates early.

Our approach. We study a budgeted active-discovery loop, ATOMICLOOP, using MATERIALS PROJECT labels as an offline oracle. Each candidate has descriptors visible to the loop and an energy-above-hull label hidden until selection. The loop trains a small MLP classifier, scores unobserved candidates, validates a batch, and repeats. We compare random search, uncertainty sampling, greedy exploitation, UCB, and DIVERSE-UCB. The method is summarized algorithmically in section 3, and the discovery curves are shown in figure 2.

Quantitative preview. At the same 2,400-candidate budget, DIVERSE-UCB found 1,933.2 stable or near-hull candidates on average, while random search found 1,364.8. This is a 41.6% increase in discoveries and a discovery acceleration factor of 1.419. The paired difference against random was +568.4 discoveries with Holm-adjusted $p = 1.05 \times 10^{-5}$. The most important negative result is also clear: uncertainty sampling found 78.2 fewer stable candidates than random, showing that uncertainty reduction and discovery maximization are different objectives.

Our contributions are:

- We formulate a leakage-controlled offline active-discovery benchmark for stable inorganic materials using real MATERIALS PROJECT energy-above-hull labels.
- We compare five selection policies under matched candidate pools, seeds, and validation budgets, with paired statistical tests against random search.
- We show that reward-oriented surrogate policies find 540–568 more stable candidates than random at a 2,400-candidate budget, while uncertainty-only sampling underperforms.
- We complement the in-domain benchmark with a GNOME transfer-ranking check and clarify why the result supports triage efficiency rather than standalone physical invention.

The rest of the paper reviews related work in AI materials discovery, defines the benchmark and acquisition policies, reports the predictive and active-discovery results, and discusses the limits of offline stability triage.

2 Related Work

Large-scale screening and active learning. MATERIALS PROJECT provides a widely used database of computed inorganic materials properties and helped establish high-throughput stability screening as a practical route to materials design Jain et al. [2013]. GNOME scaled this idea by combining graph neural networks, candidate generation, active learning, and DFT validation, reporting a large expansion of predicted stable materials Merchant et al. [2023]. Our work is smaller in scale but isolates a different question: if the oracle is fixed and the budget is limited, which policy allocates validation most efficiently?

Discovery-oriented benchmarks. MATBENCH DISCOVERY shifted evaluation from generic energy regression toward stability triage for discovery, using WBM structures and convex-hull metrics to evaluate whether models can identify stable candidates Riebesell et al. [2023]. This framing is close to our goal, but our experiment focuses on sequential selection rather than one-shot leaderboard prediction. We use discovery count, precision, recall, acceleration, confidence intervals, and paired policy comparisons to measure budgeted search behavior directly.

Generative design and LLM-guided modification. MATTERGEN uses diffusion modeling to generate inorganic crystals with stability, novelty, uniqueness, and conditioning objectives Zeni et al. [2023]. LLMATDESIGN uses language-model agents to propose interpretable structure modifications and update proposals from tool feedback Jia et al. [2024]. These approaches address candidate generation. We instead hold the candidate pool fixed and ask whether the validation loop can prioritize candidates better than trial-and-error. The two directions are complementary: generative systems still need reliable ranking, filtering, and validation policies.

Foundation atomistic models. M3GNET, CHGNET, MACE-MP, OMAT24, and UMA show that learned atomistic models can support broad materials screening, relaxation, and simulation Chen and Ong [2022], Deng et al. [2023], Batatia et al. [2024], Barroso-Luque et al. [2024], Wood et al. [2025]. These models are stronger than the lightweight surrogate used here, but they also introduce more dependencies, calibration questions, and validation cost. We use a simple MLP to

Algorithm 1 ATOMICLOOP active-discovery loop

Require: Candidate descriptors $\{\boldsymbol{x}_i\}_{i=1}^N$, hidden labels $\{y_i\}_{i=1}^N$, budget B, initial size B_0, rounds R, batch size b
Ensure: Selected set S and cumulative stable discoveries $\sum_{i \in S} y_i$
1: Select $B_0 = 400$ candidates uniformly at random and reveal their labels
2: **for** $r = 1, \ldots, R$ **do**
3: Train $p_\theta(y = 1 \mid \boldsymbol{x})$ on all revealed labels
4: Score each unselected candidate according to the acquisition policy
5: Select the next $b = 250$ candidates and reveal their labels
6: Update the observed set and cumulative discovery trace
7: **return** selected candidates and discovery trace

test whether even coarse descriptors contain enough signal for budgeted discovery; replacing it with stronger atomistic models is a direct next step.

Autonomous laboratories and validation. A-LAB demonstrated a robotic loop for inorganic synthesis, but its subsequent correction underscores the need for audited phase identification, novelty checks, and cautious interpretation Szymanski et al. [2023], Nature [2026]. Our benchmark deliberately avoids claiming new materials. It measures the triage layer only: how well an autonomous policy chooses candidates whose precomputed labels indicate thermodynamic stability.

3 Methodology

Problem formulation. Let $\mathcal{C} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^N$ be a candidate pool. The loop observes descriptors $\boldsymbol{x}_i$ for all candidates, but the binary stability label y_i is hidden until candidate i is selected for validation. We define $y_i = 1$ when energy above hull is at most 0.05 eV/atom, and $y_i = 0$ otherwise. The objective is to maximize cumulative stable discoveries under a fixed validation budget $B = 2400$.

Datasets. The primary oracle is a Hugging Face MATERIALS PROJECT snapshot with 133,420 raw rows. We excluded 3,798 rows with missing energy-above-hull labels and used 129,622 labeled rows. The external transfer check uses 139,541 GNOME R2SCAN summary rows and treats decomposition energy per atom at or below zero as the stricter on-hull label. We use GNOME near-hull rates only as a secondary diagnostic because that candidate catalog is already highly enriched.

Descriptors and leakage control. The input features include 118 element-fraction features, composition summaries such as atom count, number of elements, composition entropy, maximum and minimum element fraction, weighted atomic-number statistics, and coarse period/block summaries. For the MATERIALS PROJECT active loop we also include cell-level descriptors: volume, volume per atom, cell lengths, aspect ratio, and angles. We exclude energy-like columns, including energy per atom, formation energy, energy above hull, and decomposition energy, from all model inputs.

Surrogate model. The surrogate is a small PyTorch MLP classifier trained to estimate $p_\theta(y = 1 \mid \boldsymbol{x})$. Training uses batch size 128. Inference uses batch size 8,192. When CUDA is available, the model runs on GPU 0 with mixed precision. For the held-out predictive-signal evaluation, we train on 60,000 rows and evaluate on 20,000 rows.

Acquisition policies. Random selection is the trial-and-error baseline. Uncertainty sampling chooses candidates closest to the classifier decision boundary. Greedy selection chooses the largest predicted stability probability. UCB adds an uncertainty bonus to the predicted probability. DIVERSE-UCB starts from high-UCB candidates and spreads selection across composition space to reduce narrow exploitation. Each policy is evaluated on the same five seeds: 13, 29, 41, 53, and 67. For each seed, the candidate pool contains 25,000 MATERIALS PROJECT rows, the loop starts with 400 random observations, and the eight acquisition rounds add 2,000 candidates.

Metrics and statistical tests. The main endpoint is final cumulative stable discoveries at budget 2,400. We also report stable precision, discovery acceleration factor, and recall of stable candidates available in the candidate pool. Discovery acceleration is selected stable precision divided by the pool stable prevalence. For policy comparisons, we use paired differences against random across matched seeds. Shapiro-Wilk tests check paired differences; the selected tests were paired t-tests.

Table 1: Dataset audit for the offline discovery benchmark. Stable or near-hull means energy above hull ≤ 0.05 eV/atom for MATERIALS PROJECT. On-hull for GNOME means decomposition energy per atom ≤ 0.0.

Quantity	Value
MATERIALS PROJECT raw rows	133,420
MATERIALS PROJECT labeled rows used	129,622
Missing energy-above-hull rows excluded	3,798
Stable or near-hull rows	73,630
Stable prevalence	56.80%
GNOME R2SCAN rows	139,541
GNOME on-hull prevalence	86.96%
GNOME near-hull prevalence	98.41%

Table 2: Held-out predictive signal for the stability surrogate. Higher is better for all metrics. Best results in **bold**.

Feature set	ROC-AUC	Avg. precision	F1	Precision	Recall
Composition only	0.846	0.865	0.809	0.756	0.871
Composition plus cell	**0.874**	**0.891**	**0.828**	**0.779**	**0.883**

We report 95% confidence intervals, Cohen's d, raw tests in the artifact files, and Holm-adjusted p values across policy-vs-random comparisons.

4 Results

Data audit. table 1 reports the oracle sizes and label prevalence. The MATERIALS PROJECT pool has high stable or near-hull prevalence at 56.80%, so random search is a strong baseline rather than a sparse-discovery lower bound. The GNOME transfer set is even more enriched, with 86.96% on-hull candidates and 98.41% near-hull candidates by the thresholds used here.

The surrogate learns useful stability signal. table 2 and figure 1 show held-out predictive performance. Composition-only descriptors reach ROC-AUC 0.846 and average precision 0.865. Adding coarse cell descriptors improves ROC-AUC to 0.874, average precision to 0.891, and F1 to 0.828. This supports the first hypothesis: simple leakage-controlled descriptors contain enough signal to rank stable and near-hull candidates.

Model-guided policies discover many more stable candidates than random. table 3 reports final performance after 2,400 validations. DIVERSE-UCB finds the most stable candidates on average: 1,933.2 compared with 1,364.8 for random. Greedy and standard UCB are close behind, with 1,922.6 and 1,904.8 stable discoveries. All three reward-oriented policies increase stable precision from 0.5687 for random to 0.7937–0.8055. figure 2 shows that these gains accumulate across acquisition rounds rather than appearing only at the final budget.

The gains are statistically large across matched seeds. table 4 gives paired tests against random. DIVERSE-UCB improves final stable discoveries by +568.4 candidates with a 95% confidence interval of [524.0, 612.8] and Holm-adjusted $p = 1.05 \times 10^{-5}$. Greedy improves by +557.8 and UCB by +540.0. Uncertainty sampling is significantly worse than random by -78.2 discoveries, with Holm-adjusted $p = 0.0136$. The main positive result is therefore not just "learning helps"; it is that reward-oriented acquisition is necessary.

External transfer is positive but modest. We apply the composition-only MATERIALS PROJECT model to GNOME R2SCAN summaries because full structures were not locally mirrored. table 5 and figure 3 show that the top-ranked 100 candidates have a 0.970 on-hull rate compared with 0.8735 for random samples, an on-hull acceleration of 1.115. The gains remain positive through the top 10,000 but are modest because the GNOME set is already highly enriched.

The run is reproducible under a deterministic mini-rerun. The experiment included a same-seed repeat of the greedy policy. Both runs produced identical traces for seed 13, with

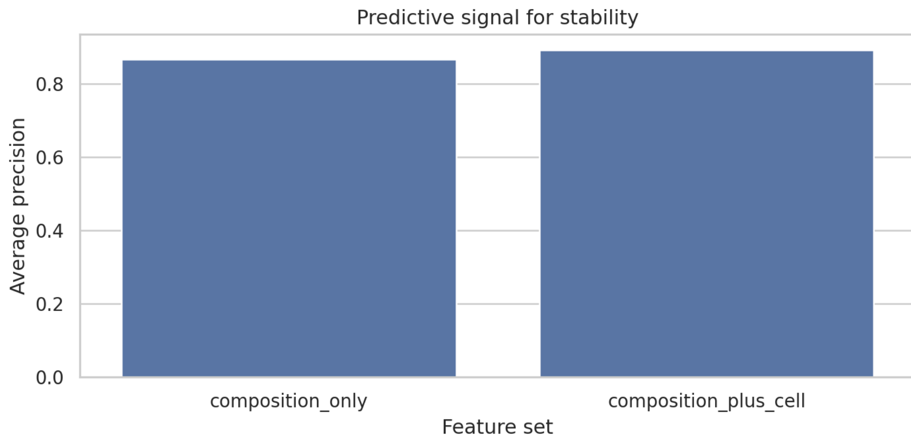


Figure 1: Predictive signal from leakage-controlled descriptors. The composition-plus-cell feature set improves the ROC and precision-recall behavior over composition-only features, showing that the surrogate has useful ranking signal before it is used inside the active-discovery loop.

Table 3: Final active-discovery performance at a fixed 2,400-candidate validation budget. Higher is better for all metrics. Best results in **bold**.

Policy	Mean stable found	Mean precision	Mean acceleration	Mean recall
DIVERSE-UCB	**1,933.2**	**0.8055**	**1.4186**	**0.1362**
Greedy	1,922.6	0.8011	1.4108	0.1354
UCB	1,904.8	0.7937	1.3977	0.1342
Random	1,364.8	0.5687	1.0015	0.0961
Uncertainty	1,286.6	0.5361	0.9442	0.0906

`same_trace=true`. This check does not prove full cross-platform determinism, but it confirms that the recorded run artifacts are internally reproducible in the configured environment.

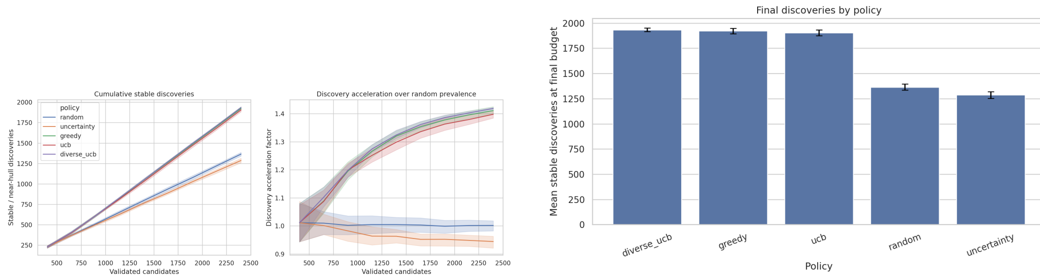
5 Discussion

What do the results show? The benchmark supports the narrow operational hypothesis behind AI-driven invention. A structured loop can use prior atomic evidence to choose better validations than random trial-and-error. The largest gain, +568.4 stable discoveries at the same 2,400-candidate budget, is practically meaningful because every avoided low-value validation can be reallocated to higher-fidelity computation or experiment.

Why did reward-oriented policies work? The supervised evaluation shows that the surrogate has real ranking signal. Greedy, UCB, and DIVERSE-UCB convert that ranking signal into high-precision selection. The small gap between DIVERSE-UCB and greedy suggests that diversity helped, but not enough to claim a strong policy-to-policy advantage with only five seeds. In this pool, exploitation already worked well because stable materials were common and the model learned a useful decision surface.

Why did uncertainty sampling fail? Uncertainty sampling selects candidates that are informative for the classifier, not necessarily candidates likely to be stable. That difference matters in autonomous discovery. A lab that spends validation budget only on ambiguous candidates can improve its map of the boundary while discovering fewer useful materials. This result argues for acquisition functions that make the objective explicit: discovery yield, novelty, cost, synthesizability, or a calibrated combination of them.

What does transfer to GNoME mean? The transfer check is deliberately conservative. The model uses composition-only descriptors because full GNoME structures were not locally mirrored. Even



(a) Discovery traces

(b) Final discoveries

Figure 2: Budgeted active discovery on MATERIALS PROJECT. Reward-oriented policies dominate random selection under the same validation budget. Uncertainty sampling underperforms random, which shows that selecting ambiguous examples is not the same as selecting high-value candidates.

Table 4: Paired policy comparisons against random across five matched seeds. Confidence intervals are for final stable-discovery differences.

Comparison	Mean difference	95% CI	Cohen's d	Holm-adjusted p
DIVERSE-UCB vs. random	**+568.4**	[524.0, 612.8]	15.90	1.05×10^{-5}
Greedy vs. random	+557.8	[514.9, 600.7]	16.16	1.05×10^{-5}
UCB vs. random	+540.0	[515.3, 564.7]	**27.19**	1.75×10^{-6}
Uncertainty vs. random	-78.2	[-129.8, -26.6]	-1.88	0.0136

under that restriction, top-100 on-hull enrichment improves from 0.8735 to 0.970. The gain is modest because GNoME is already enriched by stronger models and validation pipelines. We interpret this as weak positive evidence that MATERIALS PROJECT-trained composition priors transfer across catalogs, not as a substitute for structure-aware ranking or DFT validation.

Limitations. This is an offline benchmark, not a live autonomous laboratory. The oracle is precomputed energy-above-hull data, not new DFT, synthesis, or characterization. Stability is only one invention-relevant objective; usefulness, synthesizability, toxicity, cost, and manufacturability are not measured. The MATERIALS PROJECT pools have high stable prevalence, so future work should test lower-prevalence spaces where discovery is harder. The model is a lightweight MLP rather than CHGNET, MACE-MP, UMA, MATTERGEN, or a full MATBENCH DISCOVERY workflow. Finally, database novelty was not audited against disorder, polymorphism, or post-release entries.

Broader implications. The result supports AI-guided triage as a practical layer in materials discovery. It also argues for careful claims. A system can improve validation efficiency without proving that it has invented a new material. Stronger claims need independent novelty checks, calibrated uncertainty, high-fidelity DFT, and experimental evidence. These guardrails matter because automated systems can otherwise turn model confidence into unsupported scientific conclusions.

6 Conclusion

We presented a reproducible offline benchmark for AI-driven atomic discovery under a fixed validation budget. A lightweight surrogate-guided loop found substantially more stable or near-hull MATERIALS PROJECT candidates than random search, with DIVERSE-UCB improving final discoveries by 568.4 candidates and 41.6% at the same 2,400-candidate budget. Greedy and UCB policies also strongly outperformed random, while uncertainty sampling underperformed it.

The key takeaway is simple: structured selection can make atomic-materials validation more efficient, but prediction is only a triage layer. Future work should replace the lightweight surrogate with CHGNET, MACE-MP, OMAT24-derived models, or UMA; evaluate on harder MATBENCH DISCOVERY-style candidate spaces; add generative candidate sources such as MATTERGEN; and reserve high-fidelity DFT or experimental validation for the top candidates before making novelty or invention claims.

Table 5: GNoME transfer ranking with the composition-only MATERIALS PROJECT surrogate. Top-ranked candidates improve on-hull enrichment over random samples, while near-hull rates are nearly saturated.

Top K	Top on-hull rate	Random on-hull rate	On-hull acceleration
100	**0.970**	0.8735	**1.115**
500	0.920	0.8691	1.058
1,000	0.917	0.8702	1.055
5,000	0.937	0.8697	1.078
10,000	0.929	0.8697	1.068

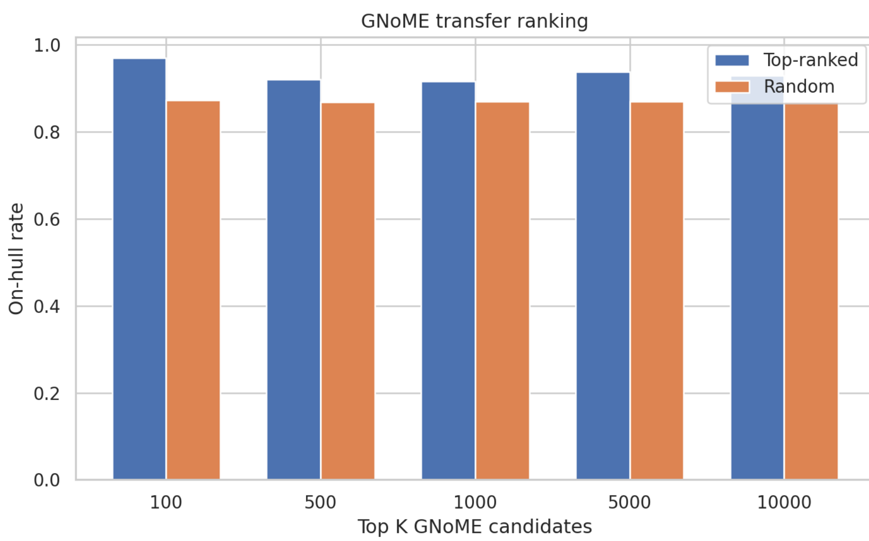


Figure 3: GNoME transfer ranking. A composition-only surrogate trained on MATERIALS PROJECT improves on-hull enrichment in the top-ranked GNoME candidates, especially at top 100. Near-hull rates are close to saturated, so the stricter on-hull rate is the more informative transfer metric.

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